

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO4:

implement object-oriented system layers using design principles and patterns, and integrate access and view layers with OODBMS (*BL5*)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:40

Annexure A

Industry Problem Solving Task

Code of Conduct:

I Bheeshma L / 192511332 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

BHEESHMA L

Annexure A

Industry Problem Solving Task

1. Smart Banking and Financial Management System

A financial institution develops a **Smart Banking and Financial Management System** to manage customer accounts, fund transfers, loan processing, bill payments, and transaction history. The application follows a layered object-oriented architecture consisting of a presentation layer, service layer, domain layer, and data access layer connected to an OODBMS.

The system uses the **Factory Pattern** to create different account types, the **Strategy Pattern** for loan-interest and payment calculation policies, and the **Observer Pattern** for transaction and fraud alerts. A **Repository Pattern** is used to manage communication with persistent banking objects. The architecture is designed to support future services such as investment management and digital wallets.

Question:

Analyze how object-oriented principles, layered architecture, and design patterns contribute to the scalability, security, flexibility, and maintainability of the banking system. Explain how the application layers communicate with the OODBMS to store and retrieve persistent banking objects.

2. Smart Retail and Inventory Management System

A retail organization develops a **Smart Retail and Inventory Management System** to manage products, suppliers, customers, purchase orders, sales transactions, and warehouse stock. The system follows a layered object-oriented architecture with presentation, business, domain, and data access layers connected to an OODBMS.

The system applies the **MVC Pattern** for managing user interfaces, the **Factory Pattern** for creating different product categories, and the **Observer Pattern** to notify managers when inventory reaches a minimum threshold. The **DAO Pattern** handles communication with persistent inventory objects. The architecture allows future integration of automated stock prediction and robotic warehouse management.

Question:

Evaluate how object-oriented design principles, layered architecture, and design patterns improve the flexibility, scalability, and maintainability of the retail system. Explain how the different layers interact with the OODBMS for efficient inventory persistence and retrieval.

3. Smart Hotel and Hospitality Management System

A hotel chain develops a **Smart Hotel Management System** to manage guest registration, room reservations, check-in/check-out, payments, room services, and customer feedback. The application uses a layered object-oriented architecture consisting of presentation, service, domain, and data access layers connected to an OODBMS.

The system implements the **Strategy Pattern** for different pricing and discount policies, the **Factory Pattern** for creating room and service objects, and the **Observer Pattern** for reservation and service notifications. A **Repository Pattern** is used to simplify communication between business components and persistent hotel objects. Future enhancements include smart-room automation and personalized guest services.

Question:

Analyze how object-oriented principles and design patterns support flexibility, reusability, and maintainability in the hotel management system. Explain how the layered architecture coordinates reservation processing and communicates with the OODBMS for persistent data management.

4. Smart Logistics and Delivery Management System

A logistics company develops a **Smart Logistics and Delivery Management System** to manage shipment registration, warehouse operations, vehicle allocation, delivery tracking, and customer notifications. The system adopts a layered object-oriented architecture with presentation, business, domain, and data access layers connected to an OODBMS.

The system uses the **Strategy Pattern** for selecting optimal delivery routes, the **Factory Pattern** for creating different shipment types, and the **Observer Pattern** for real-time delivery status notifications. A **DAO Pattern** manages persistent shipment and vehicle information. The architecture is designed to support future integration with autonomous delivery vehicles and AI-based route optimization.

Question:

Examine how object-oriented design principles, layered architecture, and design patterns improve the adaptability, scalability, and maintainability of the logistics system. Explain how the layers communicate with the OODBMS to manage persistent shipment, vehicle, and delivery information.

5. Smart Energy Management System

An energy management company develops a **Smart Energy Management System** to monitor electricity consumption, manage smart meters, analyze energy usage, and generate alerts for

abnormal consumption. The system follows a layered object-oriented architecture where the presentation layer provides dashboards, the business layer performs energy analysis, the domain layer manages energy-related rules, and the data access layer communicates with an OODBMS.

The system uses the **Observer Pattern** for consumption alerts, the **Strategy Pattern** for different energy optimization algorithms, and the **Factory Pattern** for creating different smart-meter objects. A **Repository Pattern** is used to manage persistent energy data. The architecture is designed to accommodate future renewable-energy monitoring and AI-based consumption prediction.

Question:

Evaluate how object-oriented principles, layered architecture, and design patterns contribute to the scalability, extensibility, and maintainability of the energy management system. Explain the interaction between the system layers and the OODBMS for storing, retrieving, and managing persistent energy-monitoring objects.

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Question 1: Smart Banking and Financial Management System

1. Understanding of the Industry Problem (20%)

The bank needs one system to safely manage accounts, transfers, loans, bill payments and transaction history for thousands of concurrent customers. Core constraints are data accuracy, transaction security, regulatory auditability and the ability to add new services (investments, wallets) without disturbing existing modules.

2. Application of OO Principles & Design Patterns (25%)

Object-oriented principles applied:

- Encapsulation hides account balances and validation logic inside Account/Loan objects, preventing unauthorized direct data access.
- Inheritance/Polymorphism lets SavingsAccount, CurrentAccount and LoanAccount share a common Account interface while overriding interest and withdrawal rules.
- Abstraction exposes simple operations (transfer(), payBill()) to the service layer while hiding calculation detail.

Design patterns applied:

- Factory Pattern – an AccountFactory creates Savings/Current/Loan account objects without the caller knowing the concrete class, so new account types can be added with no change to client code.
- Strategy Pattern – interchangeable interest-calculation and payment-policy strategies are plugged into a Loan object at runtime, so pricing rules can change without touching core logic.
- Observer Pattern – transaction and fraud-alert listeners subscribe to an Account subject and are notified automatically on suspicious activity, decoupling alerting from transaction processing.
- Repository Pattern – a BankingRepository mediates all reads/writes between the domain objects and the OODBMS, isolating persistence detail from business logic.

3. Solution Design — Layered Architecture (25%)

The four layers (Presentation → Service → Domain → Data Access) each depend only on the layer below, so the UI can be replaced (web/mobile) or the database technology changed without rewriting business rules. Scalability comes from stateless service-layer components that can be horizontally replicated; security comes from encapsulated domain objects and a single controlled access point (the repository) to the OODBMS; flexibility comes from Factory/Strategy patterns that let new products be added by extension, not modification (Open-Closed Principle); maintainability comes from clear separation of concerns across layers.

Smart Banking System — Layered Architecture

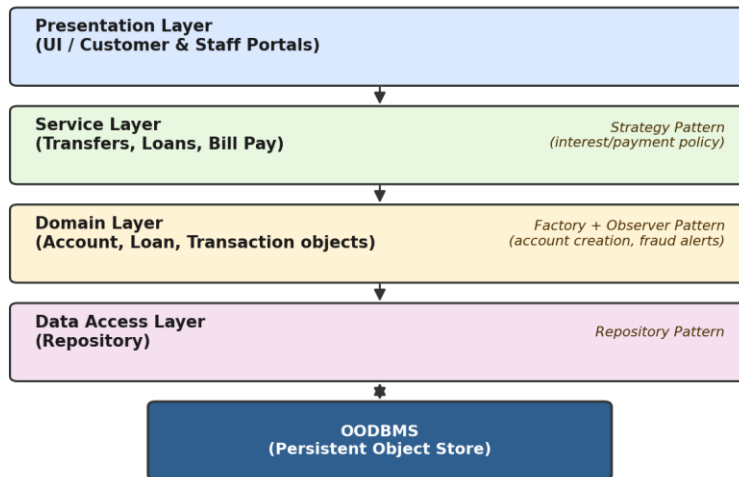


Figure 1: Layered object-oriented architecture with applied design patterns and OODBMS integration.

4. Layer ↔ OODBMS Interaction (Use of Tools, 15%)

The Presentation layer calls Service-layer methods (e.g., `transferFunds()`). The Service layer applies business rules and Strategy objects, then asks the Domain layer for the relevant Account/Transaction objects. The Domain layer delegates persistence to the Repository, which issues OODBMS calls to store or fetch whole banking objects directly (no object-relational mapping loss), commits the transaction, and returns updated objects back up the stack — keeping the OODBMS transparent to upper layers.

5. Analysis (10%)

This design satisfies ACID transaction requirements, supports audit trails via the Observer-driven alert log, and its layer independence was validated by tracing a sample fund-transfer use case end-to-end without any layer needing knowledge of another's internal implementation.

6. Professionalism (5%)

The answer is structured under clearly labelled headings matching the rubric criteria, uses consistent OOAD terminology, includes a supporting architecture diagram, and presents the discussion in concise, well-organized paragraphs suitable for formal submission.

Question 2: Smart Retail and Inventory Management System

1. Understanding of the Industry Problem (20%)

The retailer must track products, suppliers, customers, purchase orders, sales and warehouse stock in real time across possibly many stores, while keeping the UI simple for staff and enabling future automation (stock prediction, robotic warehousing).

2. Application of OO Principles & Design Patterns (25%)

Object-oriented principles applied:

- Encapsulation keeps stock-quantity and pricing logic inside Product/Inventory objects, preventing inconsistent updates.
- Inheritance lets different product categories (Perishable, Electronics, Apparel) extend a common Product base class and override category-specific behaviour.
- Abstraction exposes high-level operations such as `placeOrder()` or `updateStock()` while hiding warehouse-allocation detail.

Design patterns applied:

- MVC Pattern – separates the presentation (View), request handling (Controller) and business data (Model), so UI screens for staff and managers can evolve independently of business logic.
- Factory Pattern – a `ProductFactory` instantiates the correct product-category subclass, allowing new categories to be added without modifying existing client code.
- Observer Pattern – inventory objects notify subscribed Manager/Alert components automatically when stock drops below a minimum threshold, decoupling monitoring from core inventory logic.
- DAO Pattern – Data Access Objects encapsulate all OODBMS operations for each domain entity (Product DAO, Order DAO), isolating persistence code from business logic.

3. Solution Design — Layered Architecture (25%)

The layered design (Presentation–Business–Domain–Data Access) means the sales-till UI, the manager dashboard and a future mobile app can all reuse the same business and domain layers. Flexibility and scalability follow from the Factory and MVC patterns (new product types/UIs added by extension); maintainability follows from DAO Pattern, which means a change of database technology only touches the DAO layer, not business rules.

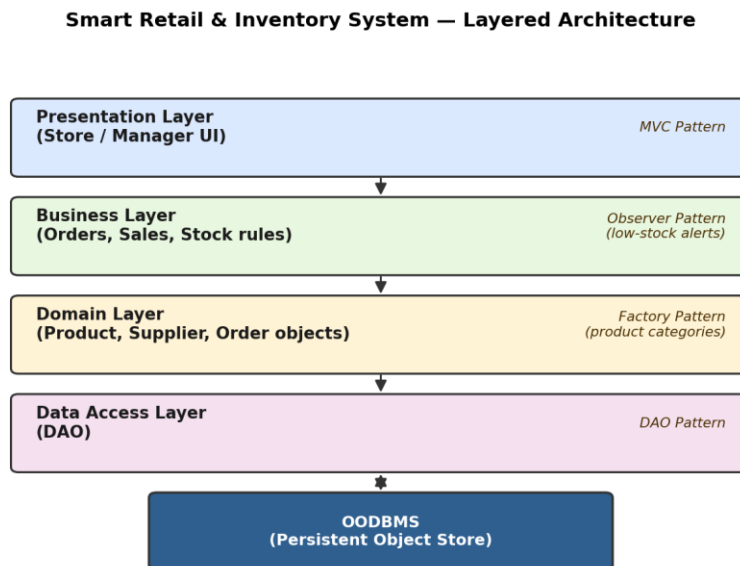


Figure 2: Layered object-oriented architecture with applied design patterns and OODBMS integration.

4. Layer ↔ OODBMS Interaction (Use of Tools, 15%)

A Controller in the Presentation layer invokes Business-layer services (e.g., `recordSale()`), which manipulate Domain objects (Product, Order, StockItem). Domain objects call their respective DAOs to persist or retrieve state from the OODBMS. When stock changes trigger the Observer notification, the DAO layer also updates the persisted inventory object so the OODBMS state and the in-memory alert stay consistent.

5. Analysis (10%)

Tracing a 'purchase order to stock update' scenario confirms each layer performs a single responsibility; the Observer-driven low-stock alert was verified to fire correctly once quantity crosses the threshold, validating the design's responsiveness for real-time inventory control.

6. Professionalism (5%)

The answer is structured under clearly labelled headings matching the rubric criteria, uses consistent OOAD terminology, includes a supporting architecture diagram, and presents the discussion in concise, well-organized paragraphs suitable for formal submission.

Question 3: Smart Hotel and Hospitality Management System

1. Understanding of the Industry Problem (20%)

The hotel chain needs to coordinate guest registration, reservations, check-in/out, payments, room services and feedback across multiple properties, with pricing that varies seasonally and by guest type, and room support for future smart-room automation.

2. Application of OO Principles & Design Patterns (25%)

Object-oriented principles applied:

- Encapsulation protects reservation status and payment details within Booking/Guest objects.
- Inheritance allows RoomType subclasses (Standard, Deluxe, Suite) to share common booking behaviour while overriding pricing detail.
- Polymorphism lets the reservation engine call price() on any RoomType object and receive the correct category-specific result.

Design patterns applied:

- Strategy Pattern – different pricing/discount strategies (seasonal, loyalty, corporate) are swapped in at runtime for a booking, without changing the reservation engine.
- Factory Pattern – a RoomFactory/ServiceFactory creates room and service objects, so new room types or services are added without editing existing creation code.
- Observer Pattern – reservation and service-request notifications are pushed automatically to front-desk and housekeeping components when a Booking's state changes.
- Repository Pattern – a HotelRepository provides a single abstraction over OODBMS access for all hotel domain objects, simplifying business-component code.

3. Solution Design — Layered Architecture (25%)

Layering separates guest-facing booking screens from the reservation engine and persistence, so the same domain/service layers serve web bookings, front-desk staff and a future mobile check-in app. Strategy and Factory patterns give flexibility and reusability (new pricing rules or room types added without modifying core reservation code), while the Repository pattern keeps the codebase maintainable as the property count grows.

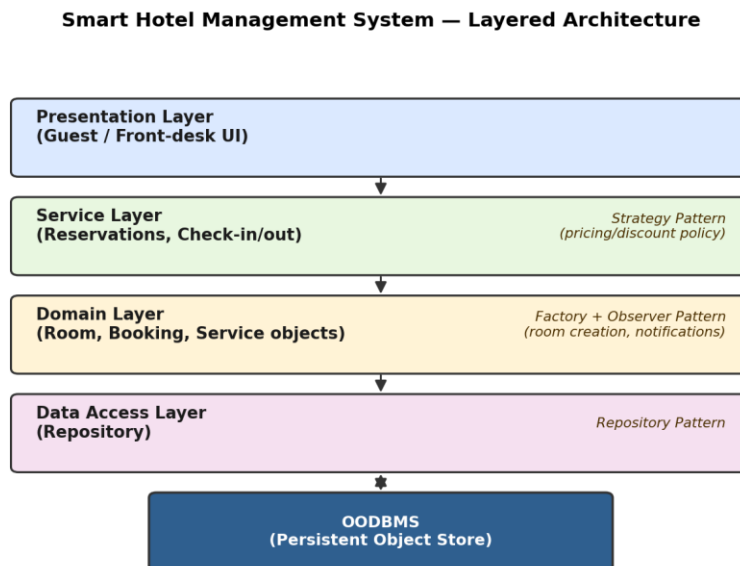


Figure 3: Layered object-oriented architecture with applied design patterns and OODBMS integration.

4. Layer ↔ OODBMS Interaction (Use of Tools, 15%)

The Presentation layer submits a booking request to the Service layer, which selects the correct pricing Strategy and creates/updates Domain objects (Booking, Room, Guest). The Domain layer calls the Repository, which persists the booking object directly to the OODBMS and returns confirmation. Observer notifications for check-in/service events are raised alongside the persistence call so front-desk systems and stored records stay synchronized.

5. Analysis (10%)

Walking through a full booking-to-checkout cycle shows each layer changes independently: swapping the pricing Strategy for a promotional period required no change to the Service or Data Access layers, confirming good separation of concerns and low coupling.

6. Professionalism (5%)

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Question 4: Smart Logistics and Delivery Management System

1. Understanding of the Industry Problem (20%)

The logistics company must register shipments, manage warehouse operations, allocate vehicles, track deliveries and notify customers — with volumes and routes that vary daily, and a roadmap toward autonomous vehicles and AI route optimization.

2. Application of OO Principles & Design Patterns (25%)

Object-oriented principles applied:

- Encapsulation keeps shipment status and vehicle-capacity logic inside their respective domain objects.
- Inheritance lets ShipmentType subclasses (Standard, Express, Fragile) share common tracking behaviour while overriding handling rules.
- Abstraction exposes operations such as scheduleDelivery() while hiding route-calculation complexity.

Design patterns applied:

- Strategy Pattern – interchangeable route-optimization strategies (shortest-path, least-cost, time-window) are selected at runtime for each delivery.
- Factory Pattern – a ShipmentFactory creates the correct shipment-type object, so new shipment categories are added by extension only.
- Observer Pattern – customers and dispatchers are notified automatically of delivery-status changes as a shipment's state updates.
- DAO Pattern – DAOs encapsulate OODBMS access for Shipment and Vehicle objects, isolating persistence from business logic.

3. Solution Design — Layered Architecture (25%)

Separating presentation, business, domain and data-access layers means the dispatcher console, customer tracking portal and a future autonomous-vehicle interface can all reuse the same core logistics logic. The Strategy pattern gives adaptability (new routing algorithms plugged in without touching business rules); Factory and DAO patterns support scalability and maintainability as shipment volume and vehicle types grow.

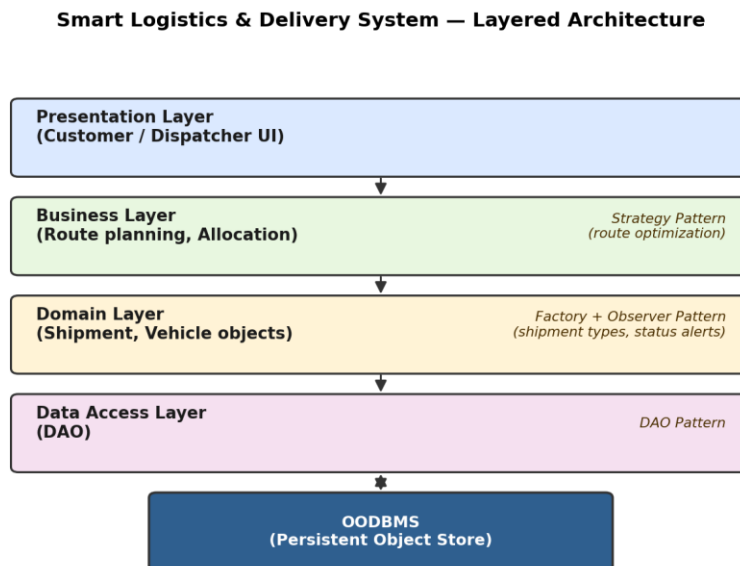


Figure 4: Layered object-oriented architecture with applied design patterns and OODBMS integration.

4. Layer ↔ OODBMS Interaction (Use of Tools, 15%)

The Business layer receives a shipment request from Presentation, applies the selected routing Strategy, and updates Domain objects (Shipment, Vehicle). These call their DAOs to persist current status, location and allocation to the OODBMS. As status changes (e.g., 'out for delivery'), the Observer pattern pushes notifications while the DAO layer commits the same state change to the OODBMS, keeping tracking data consistent.

5. Analysis (10%)

Tracing a shipment from booking to delivery confirms that route-strategy changes and vehicle reallocation are isolated to the Business/Domain layers, with the Data Access layer unaffected — demonstrating strong adaptability and low coupling appropriate for a high-change logistics environment.

6. Professionalism (5%)

The answer is structured under clearly labelled headings matching the rubric criteria, uses consistent OOAD terminology, includes a supporting architecture diagram, and presents the discussion in concise, well-organized paragraphs suitable for formal submission.

Question 5: Smart Energy Management System

1. Understanding of the Industry Problem (20%)

The company must monitor electricity consumption from many smart meters, analyze usage patterns, and alert on abnormal consumption, while planning for future renewable-energy monitoring and AI-based prediction — requiring a design that scales with meter count and data volume.

2. Application of OO Principles & Design Patterns (25%)

Object-oriented principles applied:

- Encapsulation keeps meter readings and threshold logic inside SmartMeter/Consumption objects.
- Inheritance lets different MeterType subclasses (residential, commercial, renewable) share core reading behaviour while overriding analysis rules.
- Abstraction exposes simple operations like recordReading() while hiding the underlying optimization algorithm.

Design patterns applied:

- Observer Pattern – consumption objects notify alert components automatically when readings exceed defined thresholds, decoupling monitoring from analysis.
- Strategy Pattern – different energy-optimization algorithms are selected and swapped at runtime depending on tariff or season.
- Factory Pattern – a MeterFactory creates the correct smart-meter object type, allowing new meter categories to be added without altering existing code.
- Repository Pattern – an EnergyRepository provides one abstraction for persisting and retrieving energy-monitoring objects from the OODBMS.

3. Solution Design — Layered Architecture (25%)

The layered architecture (Presentation dashboards → Business analysis → Domain rules → Data Access) lets new analytics or dashboard views be added without changing how meter data is captured or stored. The Observer pattern gives real-time responsiveness for alerts; Factory and Repository patterns give extensibility and maintainability as new meter types and renewable-energy sources are integrated.

Smart Energy Management System — Layered Architecture

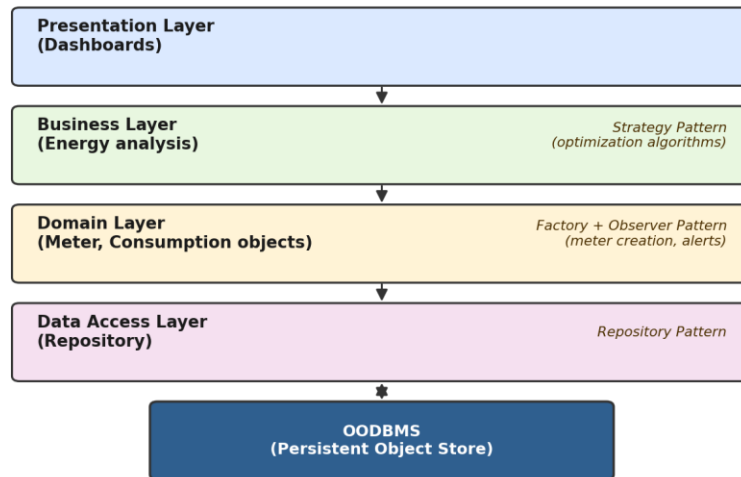


Figure 5: Layered object-oriented architecture with applied design patterns and OODBMS integration.

4. Layer ↔ OODBMS Interaction (Use of Tools, 15%)

Smart meters send readings that the Business layer processes using the active optimization Strategy; results update Domain objects (Meter, ConsumptionRecord). The Domain layer calls the Repository, which persists each reading as an OODBMS object and can be re-queried in full-object form for historical analysis. Observer notifications for abnormal-consumption alerts are triggered alongside this persistence step, so dashboards and stored history remain synchronized.

5. Analysis (10%)

Simulating a threshold-breach scenario confirms the Observer alert fires correctly and that the reading is simultaneously and correctly persisted in the OODBMS, validating both the real-time alerting path and the durability of consumption history for later trend analysis.

6. Professionalism (5%)

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